

Sick-it Learn Recreation and Implementation

CSE299-sec-5

Group-5

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Week 2 Project Report - ML Library Recreation

In Week 2 of the ML Library Recreation Project, we successfully implemented advanced regression models and preprocessing tools, significantly enhancing the library's capabilities. The key achievements include the development of **Ridge Regression with L2 regularization**, ensuring numerical stability through proper matrix operations and seamless CLI integration, and **Lasso Regression using coordinate descent optimization** for L1 regularization and automatic feature selection. Additionally, we introduced essential preprocessing scalers: **StandardScaler for z-score normalization** (mean=0, std=1) and **MinMaxScaler for range scaling** (customizable feature ranges), both fully integrated into the command-line interface for flexible data preparation workflows. **Robust input validation, comprehensive error handling, and structured logging** were maintained throughout all components, ensuring professional-grade software engineering standards and debugging capabilities. A thorough testing suite was created using Python's unittest framework to validate all new components, including model performance accuracy, preprocessing correctness, and edge case handling, enabling reliable benchmarking and systematic model comparison. The project now supports **three complete regression models** (Linear, Ridge, Lasso) and **two preprocessing scalers** with a consistent scikit-learn-compatible API, enhanced CLI usability supporting complex workflows like `python main.py --model ridge --alpha 1.0 --scaler standard`, and comprehensive documentation. Technical challenges including tuple validation errors, indentation debugging, and import resolution were successfully overcome, demonstrating strong problem-solving skills and attention to software quality. This solid architectural foundation, built on object-oriented principles with proper inheritance hierarchies and modular design, prepares the library for future expansions into classification algorithms, clustering methods, and advanced feature engineering, while the current implementation already supports real-world applications requiring regularized regression and data preprocessing capabilities.

Key Metrics:

- **3 Regression Models:** Linear, Ridge, Lasso
- **2 Preprocessing Tools:** StandardScaler, MinMaxScaler
- **Complete CLI Integration:** 8+ new command-line parameters
- **Comprehensive Testing:** 15+ unit tests with edge case coverage
- **Professional Architecture:** Consistent API, logging, and error handling

